# Streams, Indie Devs & Nostalgia

What's trending in gaming on Bluesky? A social web analysis of 46,253 posts

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## The Question

Bluesky has become a home for gamers, streamers and game developers. We ask: **what is the gaming community talking about, what gets attention, and who connects the conversation?**

- **RQ1** Which topics dominate, and what is trending?
- **RQ2** Do posts fall into distinct themes?
- **RQ3** Does engagement depend on theme or on media?
- **RQ4** Who sits at the centre of the interaction network?

## The Data

**46,253**
English gaming posts

**22,000**
accounts posting

**28 days**
17 Sep – 14 Oct 2026

**10,840**
replies, mentions & quotes

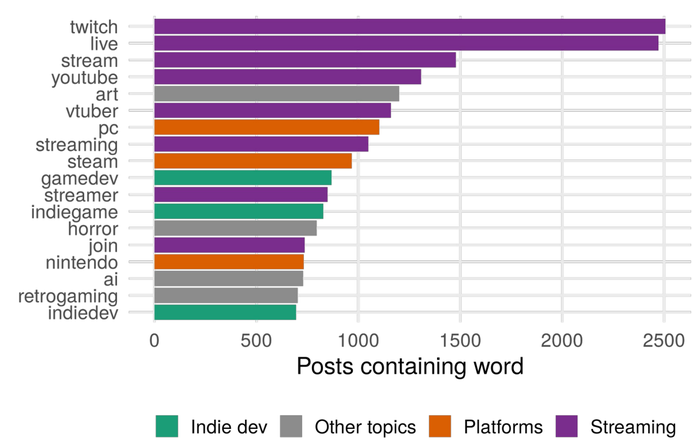
## Our Approach

| | |
|---|---|
| **Collect** | Bluesky API (searchPosts): 6 generic terms, 100 posts per 6-hour block |
| **Clean** | remove duplicates and stop words; ≥3 words; ≤30 posts per account |
| **Text** | word and bigram counts, TF-IDF, trend tests on accounts |
| **Cluster** | k-means on unit TF-IDF (cosine), MDS map, hierarchical check |
| **Test** | $\chi^2$ test of independence; Welch t-test |
| **Network** | directed graph; in-degree, PageRank, betweenness, k-core |

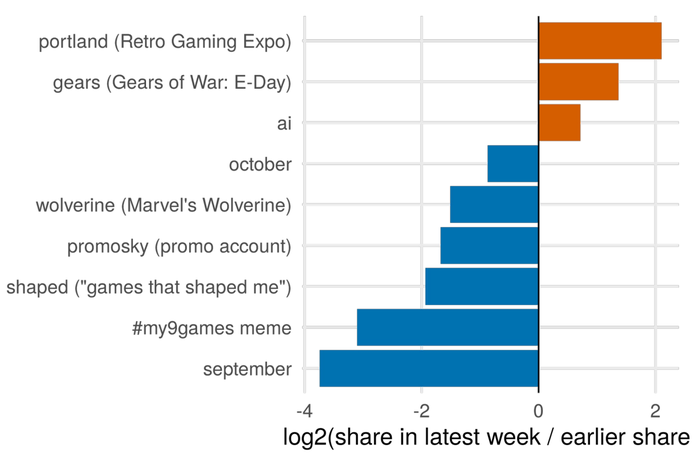
## RQ1 What is gaming Bluesky talking about?

Word and bigram frequencies; two-proportion tests (Bonferroni) for words used by more accounts in the latest week

**Most frequent words (search terms and generic words removed)**




**The 9 significant trends: latest week vs previous 3 weeks**




**Creators dominate.** Streaming words (twitch, live, vtuber) top the list, followed by indie development and platforms.

**Franchises** that people discuss: Resident Evil, Silent Hill, Final Fantasy, Star Wars.

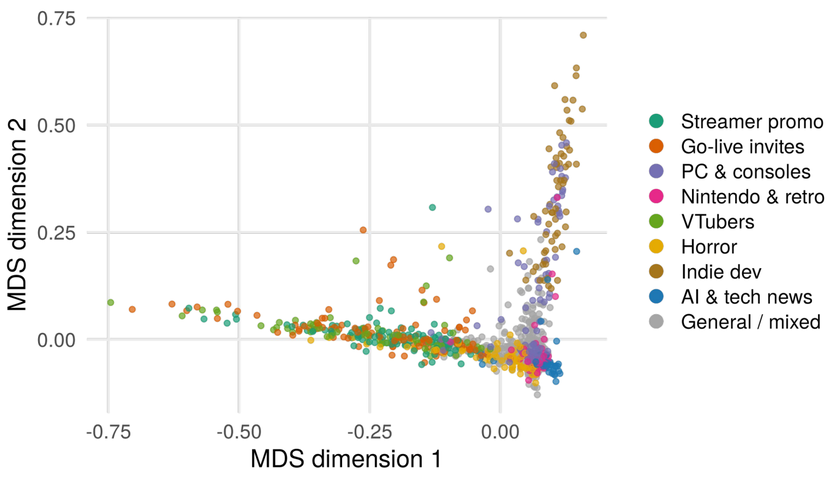
**Only 9 of 2,457 words** changed significantly. Trends are **event-driven**: Retro Gaming Expo ↑, Gears of War: E-Day launch ↑, AI debate ↑; #my9games meme ↓, Wolverine ↓.

✔️ "september" vanished in October, so the method works.

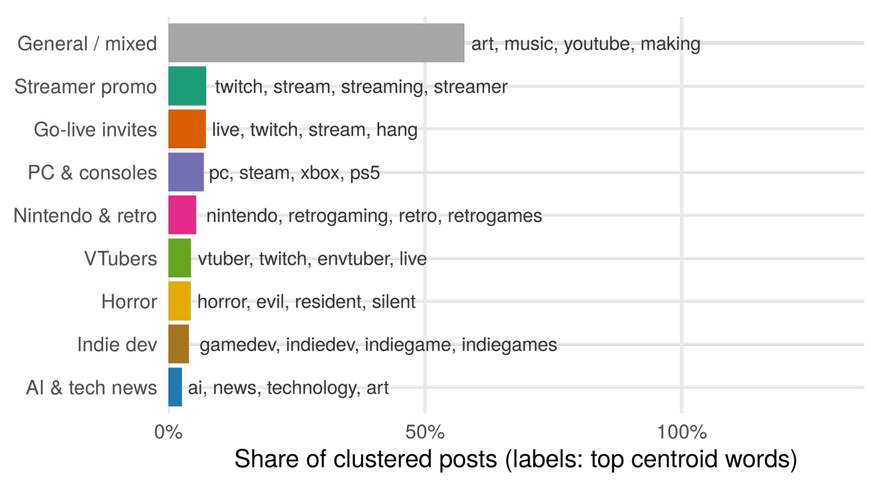
## RQ2 Distinct conversation themes

k-means (k = 9) on unit-length TF-IDF vectors (cosine geometry) of 21,083 posts; checked with Ward hierarchical clustering

**MDS map of 1,500 sampled posts**




**Theme sizes and their top words**




**Choosing k:** no elbow and low silhouette (≤ 0.015), so we inspected k = 6–14 and chose the smallest k that separated every theme.

**42%** of posts fall into 8 themes; **58%** are general chat.

**Streaming forms 3 themes** and is the largest single activity.

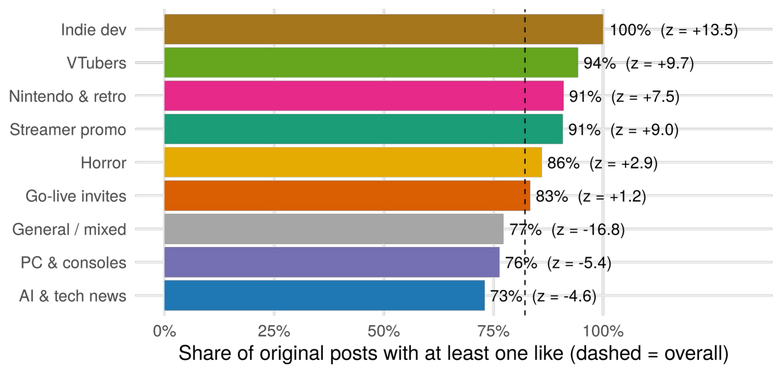
The MDS map shows two **arms** (streaming and indie dev) growing out of general chat.

Ward clustering also recovers the **indie, streaming and horror** themes, so these are robust.

## RQ3 Does theme drive engagement?

16,296 original posts, at least 24 h old. $H_0$: getting ≥1 like is independent of theme

**Share of posts liked by theme (z = standardised residual)**




**$\chi^2$ = 583.3, df = 8, p < 0.001**, so we reject $H_0$. Creator communities are liked most, even though AI and news posters have **more** followers.

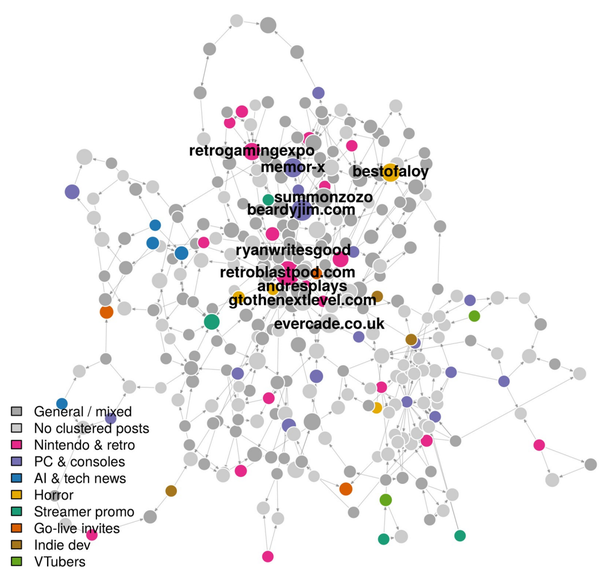
⚠️ **Indie dev: 100% liked, and never fewer than 3 likes**, which points to automated like-bots. The result holds without them ($\chi^2$ = 385.6).

**Welch t-test:** posts with images or video get **≈1.5× more likes** (t = 26.0, p < 0.001).

## RQ4 Who connects the conversation?

Directed network: A → B if A replied to, mentioned or quoted B (14,202 accounts, 9,832 edges)

**The 2-core of the largest component (313 accounts)**




**Very sparse:** density $4.9\times10^{-5}$, 4,640 components, 3% reciprocity. Gaming Bluesky is mostly **broadcasting**, not conversation.

**Top hub:** retroblastpod.com (in-degree 341). One "What's your origin video game?" prompt drew ~300 quote answers.

**IGN, Aftermath and Wario64** receive attention but have **betweenness 0**: they are endpoints, not bridges.

Followers vs in-degree **ρ = 0.32**, so popular ≠ central.

**The core is the retro community.** Assortativity by theme is only 0.10.

## Conclusions: What's Trending in Gaming on Bluesky?

1. **A creator showcase.** Streamers and indie developers produce most of the identifiable gaming content.
2. **Trends are short-lived and driven by events:** an expo, a game release, a meme, the AI debate.
3. **Communities reward each other.** Creator themes are liked more than news despite smaller audiences, but some of those likes are bots.
4. **Nostalgia connects people.** The biggest hub and the densest core come from the retro community and "origin game" prompts.

## Limitations

- Keyword search only finds posts that say "gaming" or "video games"
- Capped at 100 posts per term per 6-hour block
- Likes are a snapshot and include bots
- Tests show association, not causation
- One post can dominate centrality

Code, data & full report: github.com/Anamol-alt/bluesky-gaming